Michelson's Interferometer

Aim

To measure the wavelength of a laser, refractive index of glass slide and air.

Analysis

A beam of light passes through a beam splitter (glass slab at 45°) and reflects off from two mirors. If all the equipment was arranged ideally, then the ray diagram of the setup would look like the diagram shown below. Thus, point sources of light would appear to be two dots on the screen. On inserting a divergence lens in front of the source, we convert the point source into a plane wavefront. Thus, when the light sources reach the screen, the phases are different and are centered around the two dots. The resulting shapes that are formed due to these points have the property that the difference between all these points with each of the two dots mentioned before is a constant.

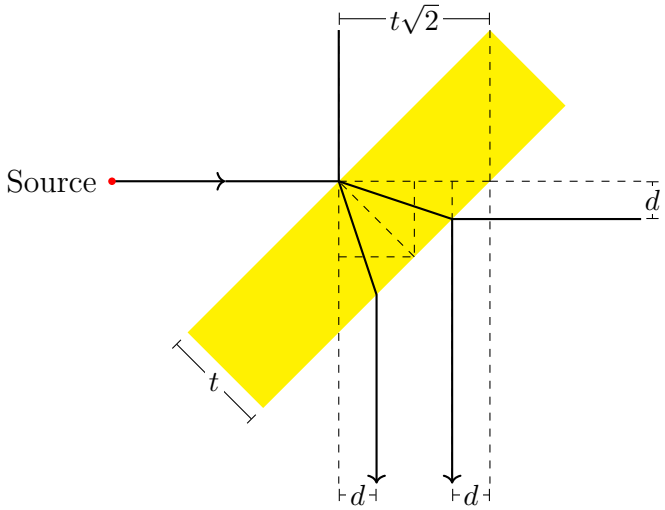


Figure 1: Here's some text

This is the definition of a hyperbola. Thus, the screen makes hyperbolic fringes with the foci as the two dots. The distance between the dots is the two times the focal length of the hyperbolic fringes (which is $t\sqrt{2}-2d$). When we adjust the settings so that both these points converge to the same point, the focal length becomes zero which results in the equation of a circle. This is how we get circular fringes and any deviation results in hyperbolic fringes.

The wavelength of the laser is calculated by $\lambda = (2d/N)\Delta$ where Δ is the caliberation constant of the micrometer and d is the micrometer reading. We first find Δ by using green laser of wavelength 532 nm. We find it to be 0.02325

Calculating values

1. For wavelength of red light

$$\lambda_r = \lambda_g \times \frac{d_r}{d_g}$$

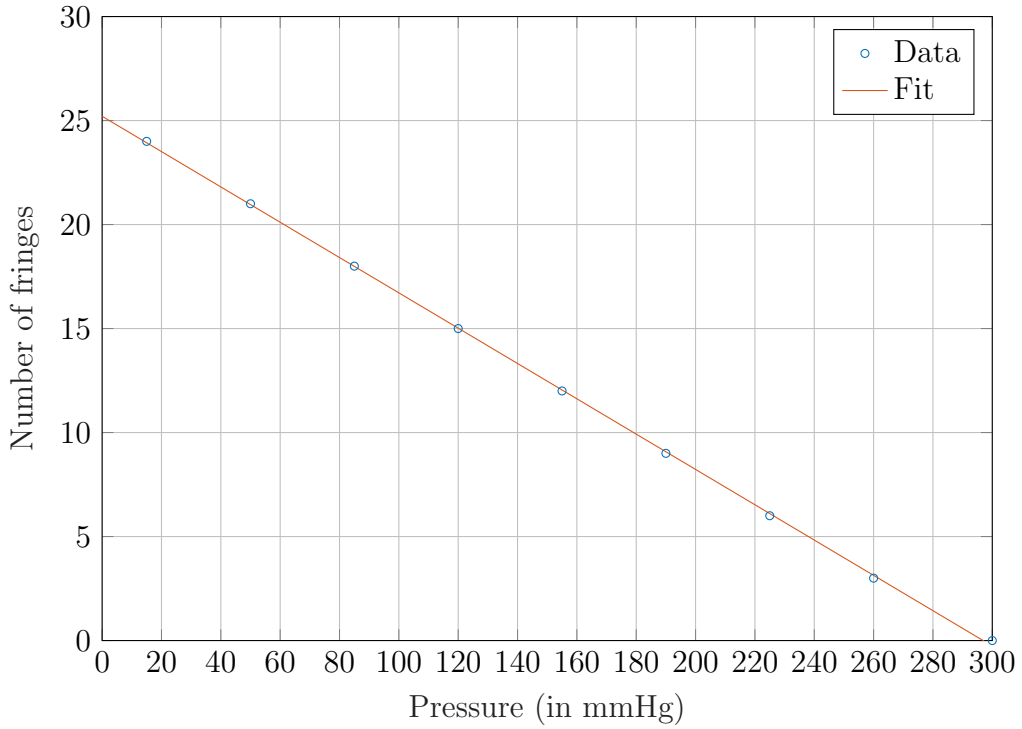
   This gives us the value for wavelength of red laser as 643.6 nm

2. For refractive index of glass slide

$$n = \frac{(2t - N\lambda)(1 - \cos\theta)}{2t(1 - \cos\theta) - N\lambda}$$

   Which gives us refractive index as 1.592

3. For refractive index of air



$$n = 1 + \frac{m}{\Delta P} \cdot \frac{\lambda}{2l} \cdot P_{atm}$$

   where l is the length of the pressure cell (8.02 cm) and m is the number change in fringes. We get the value as 1.00024

Calculating errors

If f is a function of x_is then the uncertainty in f is given by

$$\sigma_f^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \sigma_{x_i} \right)^2$$

Let $p = 1 - \cos\theta$
The uncertainties are

$$\begin{aligned} \sigma_{d_r/d_g} &= 0.034 \\ \sigma_p &= \sin\theta \cdot \sigma_\theta = 0.19 \end{aligned}$$

1. For wavelength of red light
$$\sigma_\lambda = \lambda_g \cdot \sigma_{d_r/d_g} = 18.4$$
2. For refractive index of glass slide
$$\sigma_n = \sqrt{\left(\frac{(N\lambda - 2t)(N\lambda)}{(2tp - N\lambda)^2} \cdot \sigma_p \right)^2 + \left(\frac{2tpN(1-p)}{(2tp - N\lambda)^2} \cdot \sigma_\lambda \right)^2} = 0.22$$
3. For refractive index of air
$$\sigma_n = \frac{m}{\Delta P} \cdot \frac{P_{atm}}{2l} \cdot \sigma_\lambda = 7.2 \times 10^{-6}$$

Result

We obtain the following results

1. The wavelength of red light is 643.6 ± 18.4 nm
2. The refractive index of glass slide is 1.592 ± 0.22
3. The refractive index of air is 1.00024